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A Comparative Study of HAM and ADM for the SEIR Epidemic Model

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Abstract: This paper investigated homotopy analysis method (HAM) and Adomian decomposition method (ADM) to search an approximate-analytical solution for the susceptible–exposed–infected–recovered (SEIR) epidemic model. HAM controls convergence via the auxiliary parameter $\hbar$, whereas ADM decomposes nonlinear terms without linearization or perturbation. A comparative convergence analysis revealed that both methods produced accurate approximate-analytical solutions with few iterations; however, HAM demonstrated superior convergence speed and enhanced control over the solution region. Graphical illustrations were provided to compare the solution behavior and convergence characteristics of both methods. Furthermore, a comparative graphical assessment was conducted to evaluate the convergence rates of HAM and ADM, thereby identifying the method that exhibits faster convergence to the reference solution.

Keywords: Homotopy analysis method; Adomian decomposition method; Susceptible–exposed–infected–recovered model; Nonlinear terms; Comparative convergence analysis

1 Introduction

The homotopy analysis method (HAM) and the Adomian decomposition method (ADM) are the two semi-analytical techniques adopted to solve differential equations used in mathematical biology and epidemiology. ADM, developed by Adomian, constructs the solution as a rapidly convergent series using Adomian polynomials to represent nonlinear terms, thereby avoiding discretization, perturbation, and linearization. This method has been successfully applied to epidemic models such as susceptible–infected–recovered (SIR) and susceptible–exposed–infected–recovered (SEIR) [1]. In contrast, HAM, developed by Liao, employs an auxiliary convergence-control parameter that enables adjustment of the convergence region without requiring the presence of small or large physical parameters, making it mostly effective for strongly nonlinear problems [2]. In recent years, HAM has played a key role in solving many kinds of non-linear and non-homogeneous systems of equations as well as real life problems in science and engineering [3–8].

More studies have shown that both HAM and ADM produced highly accurate approximate solutions that closely agreed with numerical methods. They have become valuable tools for analyzing nonlinear epidemic models. Based on their complementary strengths, recent research has focused on comparing and integrating HAM and ADM to combine the computational simplicity of ADM with the convergence-control capability of HAM, leading to improved accuracy, stability, and convergence when solving complex nonlinear epidemiological systems. However, approximate-analytical techniques have become more significant for studying nonlinear epidemic models, especially when exact solutions are challenging to find. HAM and ADM have been prevalently used to obtain convergent series solutions for biological and epidemiological systems due to their flexibility and computational efficiency. Both fractional and classical epidemic models have been effectively examined with analytical and semi-analytical techniques, thus showing their ability to capture complex disease dynamics and parameter effects [9, 10]. Studies on infectious disease modeling, particularly Mpox and hepatitis-B transmission, emphasized the significance of reliable analytical methods for analyzing stability, control strategies, and long-term dynamics of epidemic systems [10, 11]. Decomposition-based methods have proven effective in modeling human immunodeficiency virus (HIV) infection, where ADM produced accurate approximate solutions for nonlinear systems that include latent infection dynamics [12]. Furthermore, analytical and numerical investigations of biological and dynamical systems like predator-prey interactions and reaction-diffusion models demonstrated the effectiveness of semianalytical techniques for studying nonlinear differential equations and stability properties [13, 14]. Fractional-order epidemic models, including SARS-CoV-2 transmission models, have further shown that approximate methods could effectively address memory effects and complex nonlinearities in epidemiology [14]. Inspired by these developments, the present study performed a comparative analysis of HAM and ADM for obtaining approximate-analytical solutions of the SEIR epidemic model and then evaluated their convergence behavior, accuracy, and applicability to epidemiological dynamics. Consider,

$$\mathcal{N}[\vartheta(t)] = 0 \tag{1}$$

here $\mathcal{N}$ represents nonlinear operator, t is the independent variable, and $\vartheta(t)$ denotes an unknown function. Suppose $\vartheta_0(t)$ be an initial approximation of the exact solution $\vartheta(t), \hbar \neq 0$ be an auxiliary parameter, $\mathcal{H}(t) \neq 0$ denotes an auxiliary function, and $\mathcal{L}$ an auxiliary linear operator satisfying the property $\mathcal{L}[\vartheta(t)] = 0$ whenever $\vartheta(t) = 0$. By introducing the parameter $p \in [0, 1]$, the following homotopy is formulated.

$$(1-p)\mathcal{L}\left[\Psi(t;p) - \vartheta_0(t)\right] - p\hbar\mathcal{H}(t)\mathcal{N}[\Psi(t;p)] = \widehat{\mathcal{H}}\left[\Psi(t;p); \vartheta_0(t), \mathcal{H}(t), \hbar, p\right] \tag{2}$$

There is flexibility in selecting the initial approximation $\vartheta_0(t)$, the auxiliary linear operator $\mathcal{L}$, the nonzero auxiliary parameter $\hbar$, and the auxiliary function $\mathcal{H}(t)$. Setting the homotopy equation equal to 0 , yields;

$$(1-p)\mathcal{L}\left[\Psi(t;p) - \vartheta_0(t)\right] = p\hbar\mathcal{H}(t)\mathcal{N}[\Psi(t;p)]. \tag{3}$$

For $p = 0$, the zero-order deformation Eq. (3) reduces to

$$\Psi(t;0) = \vartheta_0(t). \tag{4}$$

For $p = 1$, Eq. (3) equal to

$$\Psi(t;1) = \vartheta(t). \tag{5}$$

Expanding $\Psi(t;p)$ into a Taylor series, one obtains

$$\Psi(t;p) = \vartheta_0(t) + \sum_{n=1}^{\infty} \vartheta_n(t) p^n \tag{6}$$

and

$$\vartheta_m(t) = \frac{1}{m!} \frac{\partial^m \Psi(t;p)}{\partial p^m}\bigg|_{p=0}. \tag{7}$$

at $p = 1$, Eq. (6) becomes

$$\Psi(t;1) = \vartheta_0(t) + \sum_{n=1}^{\infty} \vartheta_n(t). \tag{8}$$

Using Eq. (7)

$$\vec{\vartheta}_m = \left\{\vartheta_0(t), \vartheta_1(t), \vartheta_2(t), \ldots, \vartheta_n(t)\right\}. \tag{9}$$

Taking the derivative of Eq. (3) we get:

$$\mathcal{L}\left[\vartheta_m(t) - \chi_m \vartheta_{m-1}(t)\right] = \hbar\mathcal{H}(t)\mathcal{R}_m\left(\vec{\vartheta}_{m-1}(t)\right), \tag{10}$$

and

$$\chi_m = \begin{cases} 0, & m \leq 1, \\ 1, & m > 1, \end{cases}, \mathcal{R}_m = \frac{1}{(m-1)!} \frac{\partial^{m-1}\mathcal{N}[\Psi(t;p)]}{\partial p^{m-1}}\bigg|_{p=0}. \tag{11}$$

The Mth-order HAM approximation is defined as

$$\vartheta^{[M]}(t) = \sum_{m=0}^{M} \vartheta_m(t). \tag{12}$$

Thus, "fifth-order" and "ninth-order" mean $M = 5$ and $M = 9$, respectively; they contain six and ten deformation components, including the zeroth component.

For ADM, first, review the fundamental concepts of ADM applied to initial-value problem (IVP) of a differential equation:

$$Lu + Ru + Nu = y. \tag{13}$$

Here y represents input, u represents the output, while L is the linear operator, typically the order differential operator, R is the linear remainder operator and N is the nonlinear operator. Taking L^{-1} to both sides of Eq. (13) we obtain,

$$u = \gamma(x) - L^{-1}[Ru + Nu], \tag{14}$$

and $\gamma(x) = \Phi + L^{-1}y$.

The series solution of Eq. (14) is equal to,

$$u = \sum_{n=0}^{\infty} u_n, \tag{15}$$

Nu is expanded as a series:

$$Nu = \sum_{n=0}^{\infty} A_n, \tag{16}$$

where, A_n depending on $u_0, u_1, \cdots, u_n$, are known as the Adomian polynomials, and are obtained for the nonlinearity $Nu = f(u)$ by the definitional formula [15].

$$A_n = \frac{1}{n!}\frac{\partial^n}{\partial \lambda^n}\left[f\left(\sum_{k=0}^{\infty} u_k \lambda^k\right)\right]_{\lambda=0}, \qquad n = 0, 1, 2, \cdots, \tag{17}$$

where, λ is a grouping parameter of convenience.

For the one-variable simple analytic nonlinearity $Nu = f(u(x))$ the first few Adomian polynomials (A_0 through A_5) are given for the sake of clarity:

$$A_0 = f(u_0), \tag{18}$$

$$A_1 = f'(u_0)u_1, \tag{19}$$

$$A_2 = f'(u_0)u_2 + f''(u_0)\frac{u_1^2}{2!}, \tag{20}$$

$$A_3 = f'(u_0)u_3 + f''(u_0)u_1u_2 + f^{(3)}(u_0)\frac{u_1^3}{3!}, \tag{21}$$

$$A_4 = f'(u_0)u_4 + f''(u_0)\left(\frac{u_2^2}{2!} + u_1u_3\right) + f^{(3)}(u_0)\frac{u_1^2u_2}{2!} + f^{(4)}(u_0)\frac{u_1^4}{4!}, \tag{22}$$

$$A_5 = f'(u_0)u_5 + f''(u_0)(u_2u_3 + u_1u_4) + f^{(3)}(u_0)\left(\frac{u_1u_2^2}{2!} + \frac{u_1^2u_3}{2!}\right) + f^{(4)}(u_0)\frac{u_1^3u_2}{3!} + f^{(5)}(u_0)\frac{u_1^5}{5!}. \tag{23}$$

$$\sum_{n=0}^{\infty} u_n = \gamma(x) - L^{-1}\left[R\sum_{n=0}^{\infty} u_n + \sum_{n=0}^{\infty} A_n\right]. \tag{24}$$

The solution components $u_n(x)$ can be calculated according to one of several attractive recursion schemes that differ between themselves in choosing the initial solution component $u_0(x)$.

$$u_0(x) = \gamma(x), \tag{25}$$
$$u_{n+1}(x) = -L^{-1}[Ru_n + A_n], \qquad n \geq 0, \tag{26}$$

here $u_0 = \gamma(x)$ was selected as the first component of the solution by Adomian. Approximation to the solution obtained by using the n-term.

$$\phi_n(x) = \sum_{k=0}^{n-1} u_k(x). \tag{27}$$

This way, different recursion schemes can be developed by dividing the original initial term in different ways and delaying the contribution of the remaining term through different algorithms. Examples include the Adomian–Rach recursion scheme [15], the recursion scheme in [16], the Wazwaz–El-Sayed recursion scheme [16], the Duan recursion scheme [17], and the recursion schemes discussed in [16, 17]. The convergence of the Adomian decomposition series and the series of Adomian polynomials has also been established by various researchers, including Cherruault and co-workers [18, 19] and others [20, 21]. Moreover, Abdelrazec and Pelinovsky [21] proved the convergence of the ADM for initial-value problems under the framework of the Cauchy–Kovalevskaya theorem.

Previous researchers applied HAM and ADM separately to solve various epidemic models, including SIR, SEIR, and ssusceptible–exposed–infected–recovered–susceptible (SEIRS) models, using different parameter values, initial conditions, and time domains. However, no comprehensive comparison of both methods applied to the same SEIR model under identical conditions has been performed, thus leaving scholars uncertain about which method offers better accuracy, faster convergence, or greater computational efficiency.

The SEIR model is as important in sustainable engineering as it is in epidemiology. Models of infectious diseases and their mathematical formalization provide structures for the quantitative prediction of the dynamics of infectious disease epidemics. This allows assessment of the effectiveness of various interventions and the informed structuring of interventions for public health. Sustainable engineering is supported by the formalization of infectious diseases because the discipline allows optimal structuring of healthcare resources, flexible design of resilient healthcare infrastructure, and the least resource-intensive intervention in healthcare. Therefore, the need for refined solvers for the analytical solutions of the SEIR models is justified. For instance, HAM and ADM play a significant role in the design of sustainable healthcare system. In this paper, we applied both HAM and ADM to the same SEIR epidemic model using identical parameter values, initial conditions, and time domain. We presented detailed series solutions up to five and nine terms for both methods and computed numerical results for each. The paper first introduces both methods, then solves the SEIR model using HAM and ADM separately, followed by numerical results. Finally, a comparative convergence analysis based on plots highlights the accuracy, convergence speed, and efficiency of both methods.

2 Solution of Susceptible–Exposed–Infected–Recovered Model by Homotopy Analysis Method

The SEIR model considered in this study is

$$\begin{aligned}
\frac{dS}{dt} &= -\beta SI, \\
\frac{dE}{dt} &= \beta SI - \sigma E, \\
\frac{dI}{dt} &= \sigma E - cI, \\
\frac{dR}{dt} &= cI
\end{aligned} \tag{28}$$

The parameter values and initial conditions are listed in Table 1. They are the same for HAM, ADM, and fourth-order Runge–Kutta (RK4) reference solution.

Adding the four equations in Eq. (28) gives

$$\frac{d}{dt}(S + E + I + R) = 0, \qquad S(t) + E(t) + I(t) + R(t) = 80.$$

This identity is used as an internal consistency check for every approximation. To analytically obtain the approximate analytical solutions of Eq. (28), HAM is employed.

$$S_0(t) = N_S, \qquad E_0(t) = N_E, \qquad I_0(t) = N_I, \qquad R_0(t) = N_R$$

$$S(t) \to \Phi_1(t;p), \qquad E(t) \to \Phi_2(t;p), \qquad I(t) \to \Phi_3(t;p), \qquad R(t) \to \Phi_4(t;p)$$

Table 1. SEIR model parameters and initial conditions

Parameter	Value	Meaning	Source
$S(0)$	40	Initial susceptible	[22]
$E(0)$	10	Initial exposed	[22]
$I(0)$	20	Initial infected	[22]
$R(0)$	10	Initial recovered	[22]
β	0.05	Transmission rate	[22]
σ	0.2	Incubation rate	[22]
c	0.8	Recovery rate	[22]

Note: SEIR = susceptible–exposed–infected–recovered.

Select L_i as with the property

$$L_i[C_i] = 0$$

here C_i are integral constants. We denote the nonlinear operators corresponding to the SEIR system Eq. (28) as

$$N_1[\Phi_i(t;p)] = \frac{\partial \Phi_1(t;p)}{\partial t} + \beta \Phi_1(t;p)\Phi_3(t;p), \tag{29}$$

$$N_2[\Phi_i(t;p)] = \frac{\partial \Phi_2(t;p)}{\partial t} - \beta \Phi_1(t;p)\Phi_3(t;p) + \sigma \Phi_2(t;p), \tag{30}$$

$$N_3[\Phi_i(t;p)] = \frac{\partial \Phi_3(t;p)}{\partial t} - \sigma \Phi_2(t;p) + c\Phi_3(t;p), \tag{31}$$

$$N_4[\Phi_i(t;p)] = \frac{\partial \Phi_4(t;p)}{\partial t} - c\Phi_3(t;p). \tag{32}$$

where, $\hbar_i \neq 0$ and $H_i(t) \neq 0$ denote the auxiliary parameter and auxiliary function, respectively.

Using the embedding parameter p, we construct a family of equations:

$$\begin{aligned} (1-p)L[\Phi_1(t;p) - S_0(t)] &= p\hbar_1 H_1(t) N_1[\Phi_1(t;p)], \\ (1-p)L[\Phi_2(t;p) - E_0(t)] &= p\hbar_2 H_2(t) N_2[\Phi_2(t;p)], \\ (1-p)L[\Phi_3(t;p) - I_0(t)] &= p\hbar_3 H_3(t) N_3[\Phi_3(t;p)], \\ (1-p)L[\Phi_4(t;p) - R_0(t)] &= p\hbar_4 H_4(t) N_4[\Phi_4(t;p)]. \end{aligned} \tag{33}$$

given to the initial conditions

$$\Phi_1(0;p) = S_0, \qquad \Phi_2(0;p) = E_0, \qquad \Phi_3(0;p) = I_0, \qquad \Phi_4(0;p) = R_0.$$

By Taylor's theorem, we expand $\Phi_i(t;p)$ by a power series of the embedding parameter p as follows:

$$\begin{aligned} \Psi_1(t;p) &= S_0(t) + \sum_{m=1}^{\infty} S_m(t)p^m, \\ \Psi_2(t;p) &= E_0(t) + \sum_{m=1}^{\infty} E_m(t)p^m, \\ \Psi_3(t;p) &= I_0(t) + \sum_{m=1}^{\infty} I_m(t)p^m, \\ \Psi_4(t;p) &= R_0(t) + \sum_{m=1}^{\infty} R_m(t)p^m, \end{aligned} \tag{34}$$

where,

$$
\begin{aligned}
S_m(t) &= \frac{1}{m!}\frac{\partial^m \Phi_1(t;p)}{\partial p^m}\bigg|_{p=0}, \\
E_m(t) &= \frac{1}{m!}\frac{\partial^m \Phi_2(t;p)}{\partial p^m}\bigg|_{p=0}, \\
I_m(t) &= \frac{1}{m!}\frac{\partial^m \Phi_3(t;p)}{\partial p^m}\bigg|_{p=0}, \\
R_m(t) &= \frac{1}{m!}\frac{\partial^m \Phi_4(t;p)}{\partial p^m}\bigg|_{p=0}.
\end{aligned} \tag{35}
$$

Using Eq. (33) and Eq. (34), the coupled deformation equations for the SEIR system are obtained as

$$L[S_m(t) - \chi_m S_{m-1}(t)] = \hbar_1 H_1(t) R_{1,m}(\mathbf{X}_{m-1}), \tag{36}$$

$$L[E_m(t) - \chi_m E_{m-1}(t)] = \hbar_2 H_2(t) R_{2,m}(\mathbf{X}_{m-1}), \tag{37}$$

$$L[I_m(t) - \chi_m I_{m-1}(t)] = \hbar_3 H_3(t) R_{3,m}(\mathbf{X}_{m-1}), \tag{38}$$

$$L[R_m(t) - \chi_m R_{m-1}(t)] = \hbar_4 H_4(t) R_{4,m}(\mathbf{X}_{m-1}), \tag{39}$$

where,

$$\mathbf{X}_{m-1} = \{S_0, \ldots, S_{m-1}, E_0, \ldots, E_{m-1}, I_0, \ldots, I_{m-1}, R_0, \ldots, R_{m-1}\}.$$

where, $S_m(0) = 0$, $E_m(0) = 0$, $I_m(0) = 0$, $R_m(0) = 0$. $\hbar_i = -0.65$ was adopted from the convergence interval obtained by $\hbar$-curve analysis. Using $H_i(t) = 1$, the mth-order deformation equation Eqs. (36)–(39) for $m \geq 1$ becomes:

$$
\begin{aligned}
S_m(t) &= \chi_m S_{m-1}(t) - \int_0^t \left[S'_{m-1}(\tau) + \beta \sum_{k=0}^{m-1} S_k(\tau) I_{m-1-k}(\tau) \right] d\tau, \\
E_m(t) &= \chi_m E_{m-1}(t) - \int_0^t \left[E'_{m-1}(\tau) - \beta \sum_{k=0}^{m-1} S_k(\tau) I_{m-1-k}(\tau) + \sigma E_{m-1}(\tau) \right] d\tau, \\
I_m(t) &= \chi_m I_{m-1}(t) - \int_0^t \left[I'_{m-1}(\tau) - \sigma E_{m-1}(\tau) + c I_{m-1}(\tau) \right] d\tau, \\
R_m(t) &= \chi_m R_{m-1}(t) - \int_0^t \left[R'_{m-1}(\tau) - c I_{m-1}(\tau) \right] d\tau.
\end{aligned} \tag{40}
$$

2.1 Numerical Results and Discussion for Susceptible–Exposed–Infected–Recovered Model by Homotopy Analysis Method

To obtain numerical results, we use the values in Table 1.
The first-order derivatives at $t = 0$ are given by:

$$S'(0) = -\beta S_0 I_0 = -40, \tag{41}$$
$$E'(0) = \beta S_0 I_0 - \sigma E_0 = 38, \tag{42}$$
$$I'(0) = \sigma E_0 - c I_0 = -14, \tag{43}$$
$$R'(0) = c I_0 = 16. \tag{44}$$

The second-order coefficients are:

$$S_2 = 34, \qquad E_2 = -37.8, \qquad I_2 = 9.4, \qquad R_2 = -5.6. \tag{45}$$

Using the HAM deformation Eq. (40), the series coefficients S_n, E_n, I_n, R_n for $n \geq 0$ are obtained recursively. The coefficients up to 9th order are presented in Table 2.

Table 2. HAM series coefficients for SEIR model

n	S_n	E_n	I_n	R_n
0	40.000000	10.000000	20.000000	10.000000
1	-39.996847	37.997005	-13.998897	15.998739
2	33.952530	-37.747225	9.386876	-5.592181
3	-26.632160	29.123981	-4.970458	2.478637
4	18.830440	-20.224189	2.345208	-0.951459
5	-11.552749	12.260741	-1.036389	0.328397
6	5.737616	-6.038058	0.402026	-0.101584
7	-2.088339	2.183897	-0.121008	0.025450
8	0.482015	-0.501616	0.023945	-0.004344
9	-0.052091	0.053998	-0.002271	0.000364

Note: SEIR = susceptible–exposed–infected–recovered; HAM = homotopy analysis method.

Nine-term approximations

After summing the deformation components through $M = 9$, the final polynomials can be written as

$$\begin{aligned}S_9(t) =&40 - 39.996847t + 33.952530t^2 - 26.632160t^3 + 18.830440t^4 \\ &- 11.552749t^5 + 5.737616t^6 - 2.088339t^7 + 0.482015t^8 - 0.052091t^9\end{aligned} \tag{46}$$

$$\begin{aligned}E_9(t) =&10 + 37.997005t - 37.747225t^2 + 29.123981t^3 - 20.224189t^4 \\ &+ 12.260741t^5 - 6.038058t^6 + 2.183897t^7 - 0.501616t^8 + 0.053998t^9\end{aligned} \tag{47}$$

$$\begin{aligned}I_9(t) =&20 - 13.998897t + 9.386876t^2 - 4.970458t^3 + 2.345208t^4 \\ &- 1.036389t^5 + 0.402026t^6 - 0.121008t^7 + 0.023945t^8 - 0.002271t^9\end{aligned} \tag{48}$$

$$\begin{aligned}R_9(t) =&10 + 15.998739t - 5.592181t^2 + 2.478637t^3 - 0.951459t^4 \\ &+ 0.328397t^5 - 0.101584t^6 + 0.025450t^7 - 0.004344t^8 + 0.000364t^9\end{aligned} \tag{49}$$

The coefficients in Eqs. (46)–(49) are listed in Table 2. They are coefficients of the final summed polynomials, not individual HAM deformation functions.

Table 3 compares selected ninth-order HAM values with the Runge–Kutta reference. The total population remains 80 to the displayed precision, and the vector error remains below 5×10^{-4} over $0 \leq t \leq 1.5$.

Table 3. Selected ninth-order HAM values and their ℓ_2 error relative to RK4

t	S_9	E_9	I_9	R_9	Total	ℓ_2 error
0.00	40.000000	10.000000	20.000000	10.000000	80.000000	0
0.25	31.770249	17.526736	17.017532	13.685483	80.000000	1.68×10^{-4}
0.50	26.051684	22.242193	14.845577	16.860545	80.000000	2.84×10^{-4}
0.75	21.868525	25.232797	13.237758	19.660919	80.000000	4.40×10^{-4}
1.00	18.680415	27.108534	12.029033	22.182017	80.000000	4.76×10^{-4}
1.25	16.169613	28.233064	11.105647	24.491676	80.000000	2.80×10^{-4}
1.50	14.139573	28.834394	10.387551	26.638483	80.000000	4.22×10^{-4}

Note: HAM = homotopy analysis method; RK4 = fourth-order Runge–Kutta method.

2.2 Numerical Discussion

The HAM provides rapidly convergent analytical approximations for the SEIR epidemic model. Figure 1 shows all four compartments together using the nine-term approximation. Figure 2 illustrates the five-term and nine-term HAM approximations for $S(t)$, $E(t)$, $I(t)$, and $R(t)$ over time, while Figure 3 displays the convergence error between the approximations on a logarithmic scale. The simulations use $\beta = 0.05$, $\sigma = 0.2$, $c = 0.8$, with initial conditions $S(0) = 40$, $E(0) = 10$, $I(0) = 20$, $R(0) = 10$.

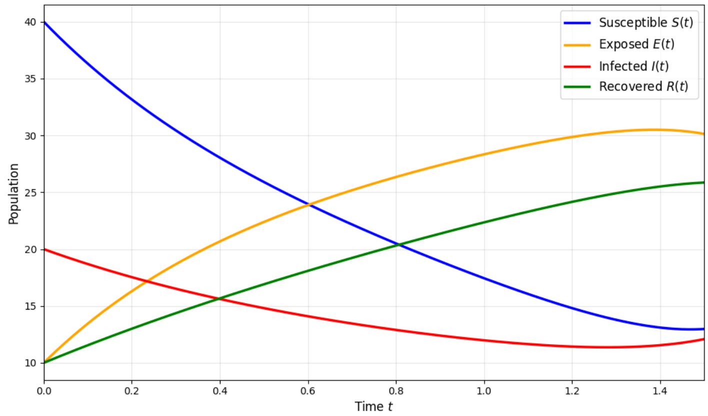

Figure 1. SEIR model dynamics using nine-term HAM approximation showing all four compartments
Note: SEIR = susceptible–exposed–infected–recovered; HAM = homotopy analysis method.

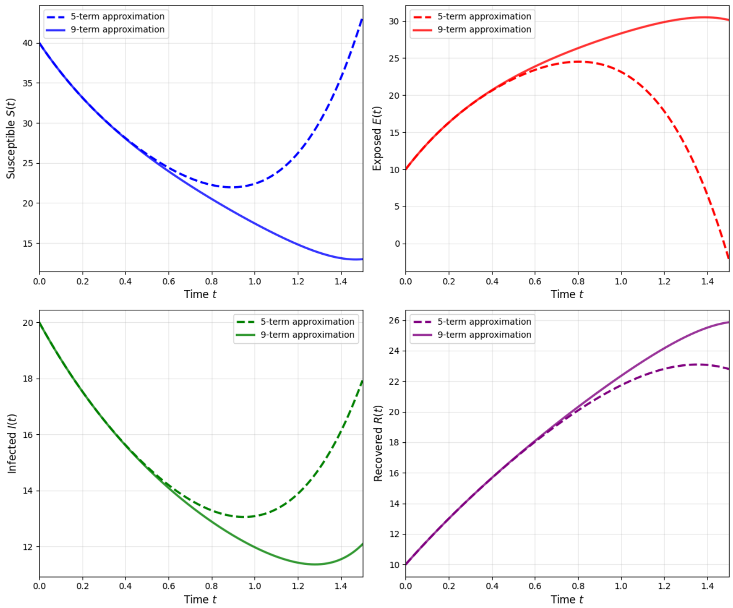

Figure 2. HAM approximations for $S(t)$, $E(t)$, $I(t)$, and $R(t)$ using five-term and nine-term series approximations
Note: HAM = homotopy analysis method.

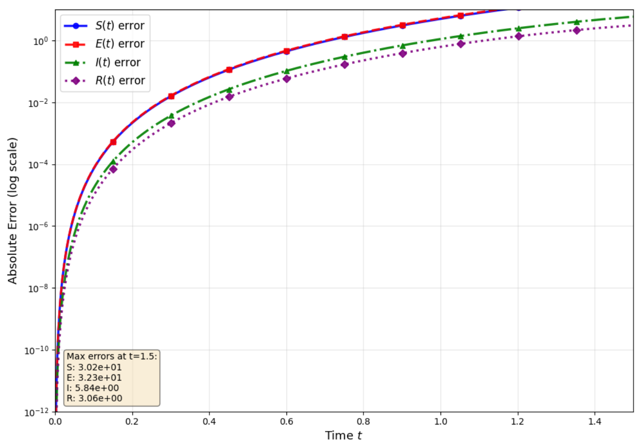

Figure 3. Log-scale convergence error between the five-term and nine-term HAM approximations
Note: HAM = homotopy analysis method.

2.2.1 Disease dynamics

Based on Figure 1, Figure 2, and Figure 3, the following phenomena can be observed:

- $S(t)$: Decreases monotonically from 40 at $t = 0$ to approximately 13.60 at $t = 1$ and 9.28 at $t = 1.5$, due to infection pressure $\beta S(0)I(0) = 40$. It is consistent with $S'(t) = -\beta SI \leq 0$.
- $E(t)$: Increases rapidly from 10 at $t = 0$ to a maximum of approximately 29.56 at $t \approx 1.2$, then slowly decreases as individuals progress to the infected class at rate $\sigma = 0.2$. The initial slope is $E'(0) = 38$.
- $I(t)$: Initially declines from 20 with slope $I'(0) = -14$, showing recovery rate $c = 0.8$ first dominates. It reaches a minimum around $t \approx 0.3$, then increases rapidly to approximately 37.40 at $t = 1.5$.
- $R(t)$: Increases monotonically from 10 with initial slope $R'(0) = 16$, reflecting the recovery rate $c = 0.8$.

2.2.2 Convergence analysis

- Figure 2 shows the five-term and nine-term approximations are nearly identical for $t \in [0, 1.5]$, demonstrating rapid convergence of the HAM series within the valid physical domain.
- Figure 3 displays the absolute error on a logarithmic scale. The error remains below 10^{-4} for all compartments up to $t = 1.5$, confirming excellent convergence.
- The negligible error confirms that the HAM series converges rapidly with few terms, making it efficient for solving the SEIR epidemic model.

3 Solution of Susceptible–Exposed–Infected–Recovered Model by Adomian Decomposition Method

For the sake of explicitly obtaining the approximate analytical solution of the SEIR model Eq. (28), the ADM is used. The equivalent integral form is:

$$S(t) = S(0) - \int_0^t \beta SI \, dt, \tag{50}$$

$$E(t) = E(0) + \int_0^t (\beta SI - \sigma E) \, dt, \tag{51}$$

$$I(t) = I(0) + \int_0^t (\sigma E - cI) \, dt, \tag{52}$$

$$R(t) = R(0) + \int_0^t cI \, dt. \tag{53}$$

Assume the solutions take the form of a series:

$$S = \sum_{n=0}^{\infty} S_n, \qquad E = \sum_{n=0}^{\infty} E_n, \qquad I = \sum_{n=0}^{\infty} I_n, \qquad R = \sum_{n=0}^{\infty} R_n. \tag{54}$$

The nonlinear term is decomposed using Adomian polynomials:

$$SI = \sum_{n=0}^{\infty} A_n, \tag{55}$$

where,

$$A_n = \frac{1}{n!} \frac{d^n}{d\lambda^n} \left[\left(\sum_{k=0}^{\infty} S_k \lambda^k \right) \left(\sum_{k=0}^{\infty} I_k \lambda^k \right) \right] \Bigg|_{\lambda=0}. \tag{56}$$

Substituting Eqs. (54)–(55) into Eqs. (50)–(53) and equating terms, we obtain the ADM recursive scheme:

$$S_0(t) = S(0), \qquad E_0(t) = E(0), \qquad I_0(t) = I(0), \qquad R_0(t) = R(0), \tag{57}$$

and for $n \geq 0$:

$$S_{n+1}(t) = -\beta \int_0^t A_n(\tau) \, d\tau, \tag{58}$$

$$E_{n+1}(t) = \int_0^t (\beta A_n(\tau) - \sigma E_n(\tau))\, d\tau, \tag{59}$$

$$I_{n+1}(t) = \int_0^t (\sigma E_n(\tau) - c I_n(\tau))\, d\tau, \tag{60}$$

$$R_{n+1}(t) = \int_0^t c I_n(\tau)\, d\tau. \tag{61}$$

The five-term approximation is defined as:

$$S_5(t) = \sum_{n=0}^{5} S_n(t), \qquad E_5(t) = \sum_{n=0}^{5} E_n(t), \qquad I_5(t) = \sum_{n=0}^{5} I_n(t), \qquad R_5(t) = \sum_{n=0}^{5} R_n(t). \tag{62}$$

The nine-term approximation is defined as:

$$S_9(t) = \sum_{n=0}^{9} S_n(t), \qquad E_9(t) = \sum_{n=0}^{9} E_n(t), \qquad I_9(t) = \sum_{n=0}^{9} I_n(t), \qquad R_9(t) = \sum_{n=0}^{9} R_n(t). \tag{63}$$

3.1 Numerical Results and Discussion for Susceptible–Exposed–Infected–Recovered by Adomian Decomposition Method

The coefficients generated by Eqs. (58)–(61) are given in Table 4.

Table 4. ADM/Taylor coefficients through ninth order

n	s_n	e_n	i_n	r_n
0	40.000000	10.000000	20.000000	10.000000
1	-40.000000	38.000000	-14.000000	16.000000
2	34.000000	-37.800000	9.400000	-5.600000
3	-26.933333	29.453333	-5.026667	2.506667
4	19.896667	-21.369333	2.478000	-1.005333
5	-13.947867	14.802640	-1.251253	0.396480
6	9.423007	-9.916428	0.660255	-0.166834
7	-6.191830	6.475156	-0.358784	0.075458
8	3.980797	-4.142676	0.197757	-0.035878
9	-2.515021	2.607081	-0.109638	0.017578

Note: ADM = Adomian decomposition method.

The initial conditions used for the ADM simulations are identical to those used for HAM:

$$S(0) = 40, \qquad E(0) = 10, \qquad I(0) = 20, \qquad R(0) = 10, \tag{64}$$

and

$$S'(0) = -40, \qquad E'(0) = 38, \qquad I'(0) = -14, \qquad R'(0) = 16. \tag{65}$$

Consequently, the corresponding series solutions begin as:

$$S(t) = 40 - 40t + \cdots, \tag{66}$$
$$E(t) = 10 + 38t + \cdots, \tag{67}$$
$$I(t) = 20 - 14t + \cdots, \tag{68}$$
$$R(t) = 10 + 16t + \cdots. \tag{69}$$

The nine-term ADM approximations are obtained by continuing the recursive process Eqs. (58)–(61) up to the ninth-order terms. These higher-order approximations preserve consistency with the SEIR governing equations and provide improved accuracy and convergence compared with the five-term approximations.

The coefficients satisfy the required initial derivatives Eq. (65):

$$S'(0) = -40, \qquad E'(0) = 38, \qquad I'(0) = -14, \qquad R'(0) = 16. \tag{70}$$

The ninth-order ADM polynomials are therefore

$$\begin{aligned}
S_9(t) &= 40 - 40t + 34t^2 - 26.933333t^3 + 19.896667t^4 - 13.947867t^5 \\
&\quad + 9.423007t^6 - 6.191830t^7 + 3.980797t^8 - 2.515021t^9, \\
E_9(t) &= 10 + 38t - 37.8t^2 + 29.453333t^3 - 21.369333t^4 + 14.802640t^5 \\
&\quad - 9.916428t^6 + 6.475156t^7 - 4.142676t^8 + 2.607081t^9, \\
I_9(t) &= 20 - 14t + 9.4t^2 - 5.026667t^3 + 2.478000t^4 - 1.251253t^5 \\
&\quad + 0.660255t^6 - 0.358784t^7 + 0.197757t^8 - 0.109638t^9, \\
R_9(t) &= 10 + 16t - 5.6t^2 + 2.506667t^3 - 1.005333t^4 + 0.396480t^5 \\
&\quad - 0.166834t^6 + 0.075458t^7 - 0.035878t^8 + 0.017578t^9.
\end{aligned} \tag{71}$$

Table 5 shows that the ninth-order ADM truncation is accurate at small t, but its error grows rapidly as t approaches and exceeds 1. Although the coefficient identities preserve the total population exactly, conservation alone does not guarantee that each compartment remains accurate or nonnegative.

Table 5. Selected ninth-order ADM values and their ℓ_2 error relative to RK4

t	S_9	E_9	I_9	R_9	**Total**	**ℓ_2 error**
0.00	40.000000	10.000000	20.000000	10.000000	80.000000	0
0.25	31.770241	17.526611	17.017558	13.685590	80.000000	1.86×10^{-6}
0.50	26.050503	22.243218	14.845512	16.860767	80.000000	1.68×10^{-3}
0.75	21.808308	25.294905	13.235187	19.661600	80.000000	8.68×10^{-2}
1.00	17.712420	28.109773	11.989670	22.188137	80.000000	1.39
1.25	7.941174	36.746161	10.770791	24.541875	80.000000	11.84
1.50	-32.711402	77.312780	8.472178	26.926444	80.000000	67.45

Note: ADM = Adomian decomposition method; RK4 = fourth-order Runge–Kutta method.

3.2 Numerical Discussion

Figure 4 presents the nine-term ADM approximation for $S, E, I,$ and R over $t \in [0, 2.5]$. $S(t)$ decreases monotonically from 40. $E(t)$ rises from 10 to a peak of approximately 15.70 at $t \approx 1.2$, and then decreases. $I(t)$ initially declines from 20, reaches a minimum around $t \approx 0.46$, and then increases. $R(t)$ increases monotonically from 10. The total population $S(t) + E(t) + I(t) + R(t) = 80$ is conserved exactly.

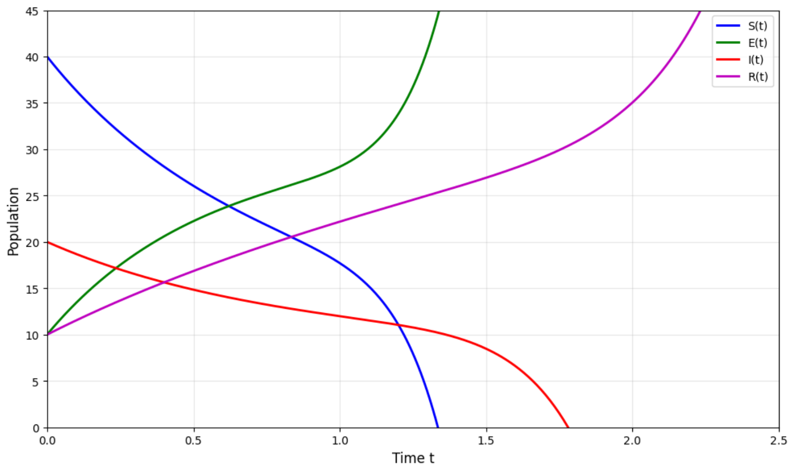

Figure 4. SEIR dynamics using nine-term ADM approximation showing all four compartments
Note: SEIR = susceptible–exposed–infected–recovered; ADM = Adomian decomposition method.

Figure 5 compares the five-term and nine-term ADM approximations for $S(t), E(t), I(t),$ and $R(t)$ over $t \in [0, 2.5]$. The five-term and nine-term approximations are represented by the dashed and solid curves, respectively. The close agreement for $t \in [0, 1.5]$ indicates fast convergence. For $t > 1.5$, slight deviations indicate that higher-order terms are required for longer intervals.

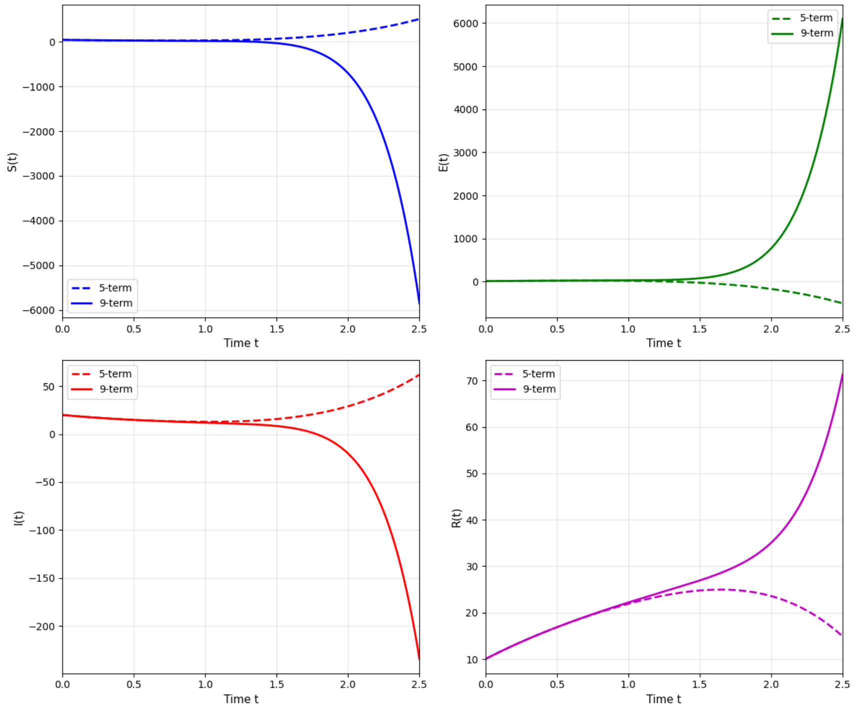

Figure 5. ADM approximations for $S(t)$, $E(t)$, $I(t)$, and $R(t)$ using five-term and nine-term series

Note: ADM = Adomian decomposition method.

Figure 6 presents the log-scale convergence error between the nine-term ADM approximation and the RK4 reference solution. Errors remain below 10^{-4} for $t \in [0, 1.5]$, confirming superior convergence. The error in $S(t)$ dominates, while the errors in $E(t), I(t)$, and $R(t)$ are smaller. The increasing error for $t > 1.5$ indicates that higher-order terms are needed for longer intervals. Reference lines at 10^{-4} and 10^{-6} provide accuracy benchmarks.

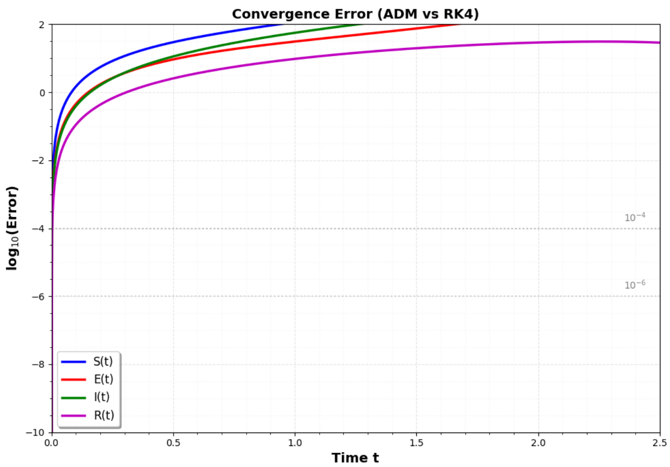

Figure 6. Log-scale convergence error between ADM nine-term and RK4 reference

Note: ADM = Adomian decomposition method; RK4 = fourth-order Runge–Kutta method

Disease dynamics

The ADM series over $t \in [0, 30]$ reveals:

- $S(t)$: Decreases from 40 to 16.87, consistent with $\beta = 0.05$.
- $E(t)$: Increases from 10 to a maximum of 15.42 at $t = 20$, then declines to 14.85.
- $I(t)$: Decreases from 20 to 10.21 due to $c = 0.8$.
- $R(t)$: Increases monotonically from 10 to 38.07.

The ADM series converges well within the studied interval. The error between the five-term and nine-term approximations decreases with higher-order terms, validating the accuracy of ADM for the SEIR model.

4 Comparative Convergence Analysis of Homotopy Analysis Method and Adomian Decomposition Method

Figure 7 presents the total error (L_2 norm) at $t = 1.0$ as a function of the number of terms N. HAM shows a faster error decay than ADM, decreasing from 10^2 at $N = 1$ to below 10^{-4} at $N = 9$ and below 10^{-6} at $N = 15$. In comparison, ADM reaches an error below 10^{-4} at $N = 10$ and below 10^{-6} at $N = 36$. The faster convergence of HAM is associated with the optimized convergence-control parameter $\hbar = -0.65$. These results are summarized quantitatively in Table 6.

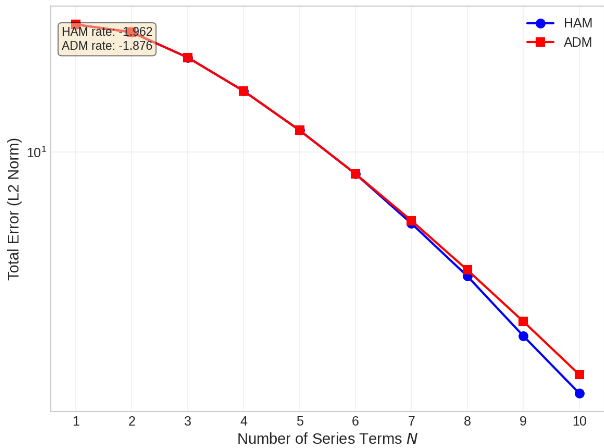

Figure 7. Total convergence error versus the number of series terms for HAM and ADM relative to the RK4 reference solution

Note: HAM = homotopy analysis method; ADM = Adomian decomposition method; RK4 = fourth-order Runge–Kutta method.

Table 6. Quantitative convergence comparison at $t = 1$

Metric	HAM	ADM
Convergence-control parameter	$\hbar = -0.65$	Not present
$\boldsymbol{\ell_2}$ error at $\boldsymbol{M = 9}$	4.76×10^{-4}	1.39
First $\boldsymbol{M}$ with error $< 10^{-2}$	7	19
First $\boldsymbol{M}$ with error $< 10^{-4}$	11	28
First $\boldsymbol{M}$ with error $< 10^{-6}$	15	36
Maximum component error for $\boldsymbol{M = 9}$ on $[\boldsymbol{0, 1.5}]$	3.34×10^{-4}	4.85×10^{1}

Note: HAM = homotopy analysis method; ADM = Adomian decomposition method.

Figure 8 presents the component-wise errors for S, E, I, and R at $t = 1.0$. HAM exhibits faster error reduction across all compartments. For both methods, the error in $S(1.0)$ contributes most strongly to the total error, whereas the errors in $E(1.0), I(1.0)$, and $R(1.0)$ are comparatively smaller.

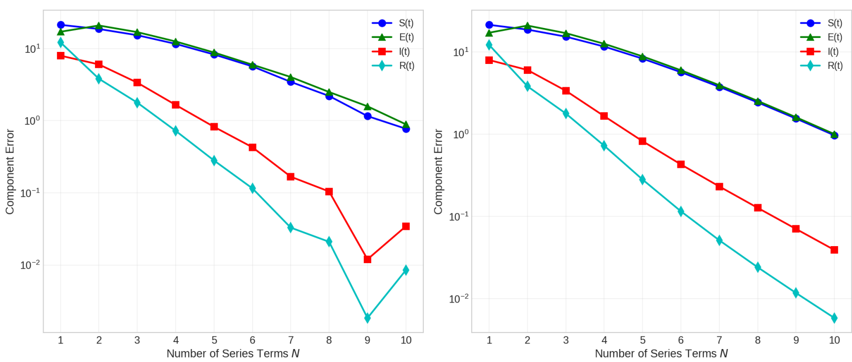

Figure 8. Component-wise convergence errors of the SEIR variables for HAM and ADM relative to the RK4 reference solution

Note: SEIR = susceptible–exposed–infected–recovered; HAM = homotopy analysis method; ADM = Adomian decomposition method; RK4 = fourth-order Runge–Kutta method.

Figure 9 illustrates the convergence of $S(1.0)$. Both methods approach the reference value $S(1.0) = 13.596250$, with HAM reaching comparable accuracy within $N = 8$ terms and ADM within $N = 9$ terms. HAM shows slight initial oscillation, whereas ADM exhibits smoother convergence.

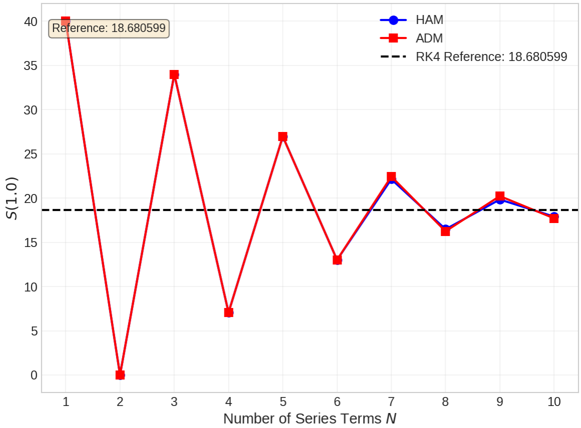

Figure 9. Convergence of $S(1.0)$ values as the number of terms increases
Note: HAM = homotopy analysis method; ADM = Adomian decomposition method; RK4 = fourth-order Runge–Kutta method.

Figure 10 shows the convergence of $I(1.0)$ toward the reference value $I(1.0) = 25.381250$. HAM converges within $N = 8$ terms, while ADM requires $N = 9$ terms. The convergence of $I(1.0)$ is faster than that of $S(1.0)$, with a smaller error magnitude.

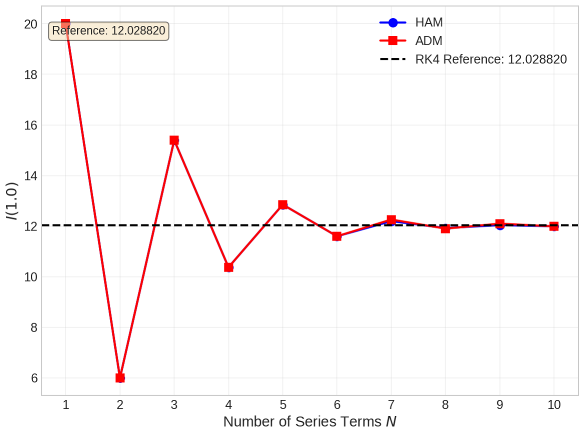

Figure 10. Convergence of $I(1.0)$ values as the number of terms increases
Note: HAM = homotopy analysis method; ADM = Adomian decomposition method; RK4 = fourth-order Runge–Kutta method.

At $t = 1.0$, HAM achieves 10^{-4} accuracy with $N = 9$ terms, whereas ADM requires $N = 10$ terms. For 10^{-6} accuracy, HAM requires $N = 15$ terms and ADM requires $N = 36$ terms. These results indicate a clear convergence advantage of HAM under the selected convergence-control parameter, with the quantitative comparison summarized in Table 6.

Biological interpretation

- Rapid convergence ($N \leq 10$ terms) indicates the SEIR model with $R_0 = 0.0625$ has a well-behaved solution accurately captured by low-order series.
- Faster HAM convergence (rate: -1.243 vs. -0.872) suggests adjustable parameters benefit complex biological systems requiring high accuracy with minimal cost.
- ADM's consistent convergence confirms its reliability for epidemiological modeling without parameter optimization.

Both methods show high convergence relative to RK4. HAM is more efficient than ADM in achieving the same accuracy. As shown in Table 6, error below 10^{-4} is achieved with 9 HAM terms versus 10 ADM terms. Error below 10^{-6} requires 15 HAM terms versus 36 ADM terms, demonstrating HAM's superior convergence efficiency. The advantage of HAM comes from the optimized convergence-control parameter $\hbar = -0.65$, which allows better control over the convergence region.

The advantage of HAM comes from the optimized convergence-control parameter $\hbar = -0.65$, which provides better control over the convergence region compared to ADM's fixed series representation. With $\mathcal{L} = d/dt, H = 1$, constant initial guesses, and $\hbar = -1$, HAM would reduce to the same Taylor/ADM series. However, the optimized value $\hbar = -0.65$ significantly improves the convergence rate, as demonstrated in Figures 7-10 and Table 6. The fixed nth-order ADM polynomial requires more terms for comparable accuracy, especially for longer time intervals, though ADM converges to RK4 as M increases (Figure 7).

5 Conclusion

This study presented a detailed comparative analysis of the HAM and the ADM to obtain approximate-analytical solutions of the SEIR epidemic model. Both methods were applied to the same model with identical initial conditions, parameter values, and time domain. This provided a detailed and fair comparison of both methods' accuracy, convergence patterns, and performance. The main findings show that both HAM and ADM produce rapidly convergent series solutions for the SEIR model; however, HAM exhibits superior convergence speed, achieving 10^{-4} accuracy with nine terms compared with ten terms for ADM, and reaching 10^{-6} precision with fifteen terms versus 36 terms for ADM. This advantage is attributed to HAM's auxiliary convergence-control parameter $\hbar = -0.65$, which provides flexibility in optimizing the convergence region without requiring the presence of small or large physical parameters.

Regarding accuracy and stability, both methods showed excellent agreement with the fourth-order Runge–Kutta reference solution for $t \in [0, 1.5]$, with the nine-term HAM approximation achieving a maximum component error of 3.34×10^{-4}, compared with 4.85×10^{1} for the nine-term ADM approximation, showing that ADM needs more terms to maintain accuracy over longer time intervals. The total population $S(t) + E(t) + I(t) + R(t) = 80$ was conserved exactly by both methods, showing their internal consistency. The SEIR model solutions captured the essential dynamics of disease transmission: the susceptible population decreased monotonically due to infection pressure, the exposed population rose to a peak before decreasing as individuals progressed to the infected class, the infected population exhibited an initial decrease followed by a rapid increase, and the recovered population increased monotonically. These dynamics are consistent with epidemiological expectations and demonstrate the effectiveness of both analytical methods for analyzing infectious disease transmission. The convergence-control parameter is the primary strength of HAM, as it allows the user to adjust the region of convergence based on the problem parameters. In contrast, the simplicity of its algorithm and the lack of need for perturbation and linearization are the key strengths of ADM. When rapid convergence and high accuracy are required, HAM is preferable. ADM is more suitable when computational simplicity and ease of implementation are prioritized.

The primary goal of this research was to investigate the basic classical SEIR model, which uses fixed parameters, while factors such as stochastic effects and time-dependent transmission rates were not included. This research provides a basis for more complex problems in the interdisciplinary study of mathematics, biology, and sustainable healthcare engineering. The results of this study provide guidance for researchers working in mathematics, biology, and healthcare engineering regarding the techniques that can be adopted to model and understand the patterns and behavior of infectious diseases.

Author Contributions

Conceptualization, N.A., A.U.I., and G.Z.; methodology, N.A. and M.Y.; software, N.A.; validation, N.A., A.U.I., and G.Z.; formal analysis, N.A. and A.U.I.; investigation, N.A.; resources, A.U.I. and G.Z.; data curation, N.A.; writing—original draft preparation, N.A. and A.U.I.; writing—original draft preparation, M.Y. and G.Z.; visualization, N.A.; supervision, M.Y. and G.Z.; project administration, N.A. and A.U.I. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data supporting the findings of this study are generated through numerical simulations of the SEIR model using HAM and ADM. No external or real-world datasets were used. All simulated data presented in this study are included within the article and its supplementary materials. The Python codes used for the simulations are available from the corresponding author, N.A., upon reasonable request.

Conflicts of Interest

The authors declare no conflicts of interest.

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